Lab 36 — Pandas Plotting (Data Visualisation)

Aim

To visualise a DataFrame's data using Pandas' built-in plotting functions by generating a histogram and bar chart.

Algorithm / Procedure

1. Import Pandas and Matplotlib.
2. Create a DataFrame containing student names and marks.
3. Generate a histogram to show the marks distribution.
4. Save the histogram as marks_hist.png.
5. Generate a bar chart showing marks of each student.
6. Save the bar chart as marks_bar.png.
7. Display the confirmation message.

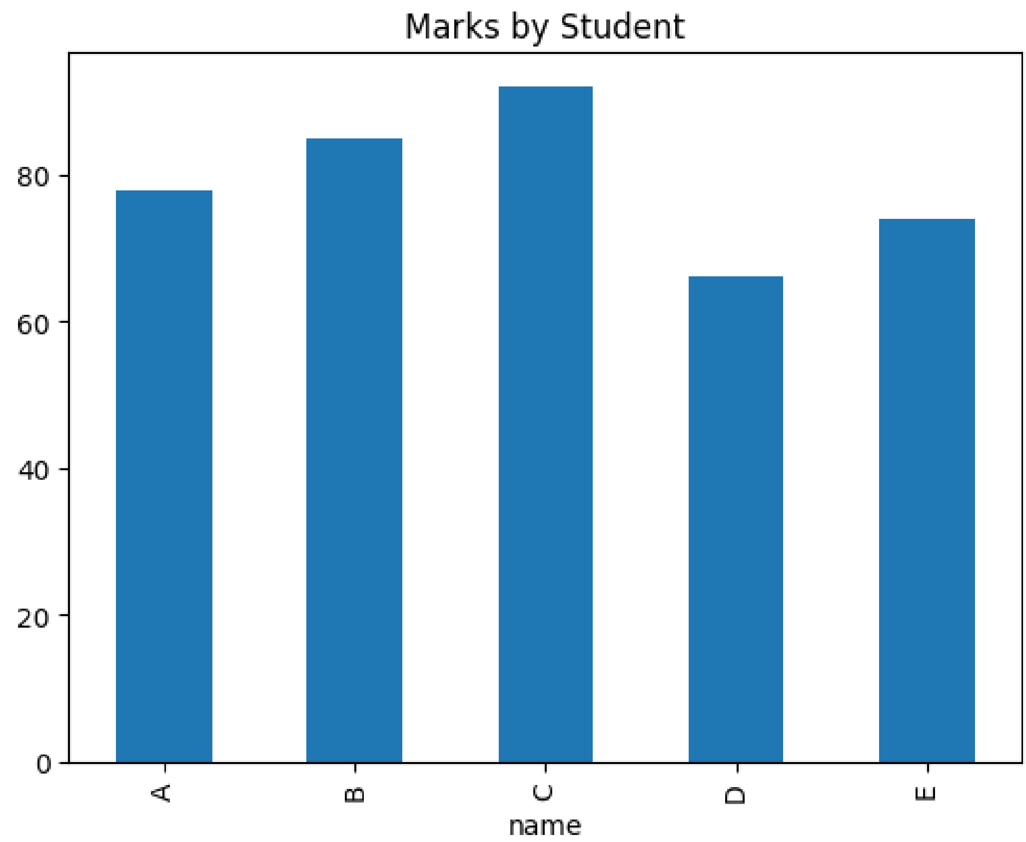
Program

```
import pandas as pd

import matplotlib.pyplot as plt

df=pd.DataFrame({'name':['A','B','C','D','E'],'marks':[78,85,92,66,74]})

df['marks'].plot(kind='hist',title='Marks Distribution',bins=5)

plt.savefig('marks_hist.png')

plt.clf()

df.plot(x='name',y='marks',kind='bar',title='Marks by Student',legend=False)

plt.savefig('marks_bar.png')

print("Plots saved: marks_hist.png, marks_bar.png")
```

Output

Plots saved: marks_hist.png, marks_bar.png

<Figure size 640x480 with 0 Axes>



Result

Pandas plotting functions were successfully used to generate a histogram and bar chart for visualising student marks.